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Evolution of Arthropod Neural Patterning

It is now known that many genes and gene networks involved in development have been conserved throughout evolution and have retained their functions in a diverse range of organisms. Coding sequences and expression patterns of the Hox genes, segmentation genes and genes involved in limb development have been compared in a number of different species of vertebrates and invertebrates. These studies corroborate older theoretical work that suggested that morphological, and likely physiological and behavioral, diversity is accompanied by two inter-dependent types of genomic change: the generation of ‘new’ genes by gene duplication and by ‘tinkering’ with old genes, via changes in their regulation (Jacob, 1977). Duplicated genes, freed from selective constraint, can diversify and adopt new functions. Gene duplications also allow for more sophisticated regulation of expression. Increased complexity of expression requires the modification of regulatory sequences, presumed to be created by an increasingly modular arrangement of

enhancer elements.

Task Description:

The recent emergence of comparative molecular embryology has largely focused on the organization of Hox clusters and on Hox gene expression. Hox genes regulate the basic body plan of most animals (Akam, 1995; Holland and Garcia-Fernandez, 1996), thus significant alterations to Hox gene number or regulation will likely result in dramatic phenotypes associated with reduction in organism fitness and thus not be fixed in a population. Comparative analyses of Hox genes may then provide a restricted view of the types of modifications that can lead to morphological diversity within the animal kingdom. We hypothesize that a comparative analysis of other developmental genes less prone to evolutionary restrictions will provide a complementary view of the genetic basis of morphological diversity. In this vein, we propose to carry out a comparative analysis of achaete-scute gene number, regulation and organization in arthropods. achaete-scute gene expression largely determines the nervous system pattern in invertebrates. achaete-scute genes, like Hox genes, are located in a complex and encode transcription factors. However, alterations to achaete-scute gene number or regulation, unlike with Hox genes, will display localized effects on nervous system patterning and not global effects on body plan organization. The genes of the achaete-scute complex should then provide an evolutionarily more plastic model system than the Hox genes to uncover the genetic basis of morphological diversity. To determine the functional significance of the vnd, ind and msh genes during Tribolium central nervous system (CNS) development. Organisms use their nervous system to communicate with and to respond to their environment. Evolutionary changes in nervous system pattern and physiology are key factors that allow organisms adapt to different ecological niches. In order to understand the molecular factors that underlie evolutionary modifications to nervous system pattern we must understand the molecular differences (and similarities) between closely related organisms with differences in their nervous system pattern. Our research focuses on understanding the molecular and genetic differences (and similarities) between central neurogenesis in Tribolium and Drosophila.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

A) Determine the complement of achaete-scute genes in four arthropod species (Tribolium, Thermobia, Triops and Lithobius) via PCR and library screening methods. Using the sequences identified, estimate the origin of unique achaete-scute family members within the arthropod lineage via phylogenetic analysis.
B) Determine the expression pattern of achaete-scute genes during CNS development in Tribolium, Thermobia, Triops and Lithobius via standard gene expression analysis. These analyses will provide a detailed view of the cellular processes that select cells as neural precursors throughout the arthropod phylum.

C) Determine the genomic organization of the achaete-scute genes in select arthropods. Clone and sequence the achaete-scute complex from D. virilis, Tribolium, and Thermobia.

D) Identify CNS regulatory regions in Tribolium and Thermobia.

Significance: Most comparative embryological work has focused on the analysis of Hox clusters and Hox gene expression in vertebrates and invertebrates. The major theme that emerges from this work implicates cis-regulatory changes as the primary mechanisms through which alterations to Hox genes led to morphological variation of the body pattern within vertebrates and invertebrates. Since Hox genes regulate the basic body plan of most animals, significant alterations to Hox gene number or regulation will likely result in dramatic phenotypes associated with reduction in organism fitness and thus not be fixed in a population. Thus, comparative analyses of Hox genes may provide a limited view of the types of modifications that can lead to morphological diversity within the animal kingdom. In order to obtain a more expansive view of the genetic basis of morphological diversity we focus our proposal on a comparative analysis of achaete-scute gene number, regulation and organization in arthropods. Alterations to achaete-scute gene number or regulation, unlike with Hox genes, will display localized effects on nervous system patterning and not global effects on body plan organization. The genes of the achaete-scute complex are then likely to provide an evolutionarily more plastic model system than the Hox genes to uncover the genetic basis of morphological diversity.

Task Progress:

Description of scientific goals The goals of our research project were (i) to complete a comprehensive analysis of achaete-scute gene expression and function during CNS development in the red flour beetle Tribolium, (ii) to determine the roles of the vnd, ind and msh genes in regulating regulate achaete-scute gene expression and neural precursor formation along the DV axix of the Tribolium CNS, and (iii) to explore the expression and function of achaete-scute genes in basal insects and crustacea.
Summary of research accomplishments A) To complete a comprehensive analysis of achaete-scute gene expression and function during CNS development in Tribolium.

Summary of Work The study of achaete-scute (ac/sc) genes has become a paradigm to understand the evolution and development of the arthropod nervous system. We identified and characterized the ac/sc genes in the coleopteran insect species Tribolium castaneum. We identified two Tribolium ac/sc genes – achaete-scute homolog (Tc-ASH) a proneural gene and asense (Tc-ase) a neural precursor gene that reside in a gene complex. Focusing on the embryonic central nervous system we found that Tc-ASH is expressed in all neural precursors and the proneural clusters from which they segregate. Through RNAi and misexpression studies we showed that Tc-ASH is necessary for neural precursor formation in Tribolium and sufficient for neural precursor formation in Drosophila. Comparison of the function of the Drosophila and Tribolium proneural ac/sc genes suggests that in the Drosophila lineage these genes have maintained their ancestral function in neural precursor formation and have acquired a new role in the fate specification of individual neural precursors. Furthermore, we found that Tc-ase is expressed in all neural precursors suggesting an important and conserved role for asense genes in insect nervous system development. Our analysis of the Tribolium ac/sc genes indicates significant plasticity in gene number, expression and function, and implicates these modifications in the evolution of arthropod neural development.

B) To determine the roles of the vnd, ind and msh genes in regulating regulate achaete-scute gene expression and neural precursor formation in the Tribolium CNS.

Summary of work: The Drosophila columnar genes are key regulators of neural precursor formation and patterning along the dorsal-ventral axis of the developing CNS and include ventral nerve cord defective (vnd), intermediate nerve cord defective (ind), muscle segment homeodomain (msh) and Epidermal growth factor receptor (Egfr). To investigate the evolution of neural pattern formation we identified and determined the expression patterns of Tribolium vnd, ind and

	msh, and found they are expressed in the medial, intermediate and lateral columns of the developing CNS, respectively, in patterns similar, but not identical, to their Drosophila orthologs. The pattern of Egfr activity suggests that the genetic regulatory mechanisms that initiate Tc-vnd expression are similar in Drosophila and Tribolium, whereas those that initiate Tc-ind have diverged. RNAi analyses of gene function showed that Tc-vnd and Tc-ind promote the formation of medial and intermediate column neural precursors and that vnd-mediated repression of ind establishes the boundary between the medial and intermediate columns. These data suggest that columnar gene expression and function underlie neural pattern formation in Drosophila, Tribolium, and potentially all insects, but that subtle spatiotemporal differences in expression of these genes may produce species-specific morphological differences. C) To explore the expression and function of achaete-scute genes in basal insects and crustacea. Summary of this work: Prior to this work achaete-scute (ac/sc) gene expression had been investigated in insects, chelicerates, and myriapods. This work identified two ac/sc genes from the branchiopod crustacean Triops longicaudatus. Triops longicaudatus achaete-scute homologs1 and 2 (Tl-ASH1 and Tl-ASH2) exhibit dynamic and distinct expression profiles during Triops neurogenesis. Tl-ASH1 expression initiates in nearly all cells of the neurogenic region and subsequently in clusters of cells evenly spaced along the length of the developing limbs. In contrast, Tl-ASH2 initiates expression after Tl-ASH1. In the CNS, only a subset of Tl-ASH1 cells appears to express Tl-ASH2. Similarly, in the PNS individual Tl-ASH2 positive cells appear to arise from the clusters of Tl-ASH1 expressing cells. Shortly after activating Tl-ASH2 expression, these cells enlarge and divide. The expression dynamics of ac/sc genes in Triops parallel those observed in insects and contrasts with those found in chelicerates and myriapods.
Bibliography Type:	Description: (Last Updated: 11/23/2005)
Abstracts for Journals and Proceedings	Wheeler, S.R.; Carrico, M.L.; Wilson, B.A. and Skeath J.B. "The Tribolium columnar genes reveal conservation and plasticity in neural precursor patterning along the embryonic dorsal-ventral axis." The Drosophila columnar genes are key regulators of neural precursor formation and patterning along the dorsal-ventral axis of the developing CNS and include ventral nerve cord defective (vnd), intermediate nerve cord defective (ind), muscle segment homeodomain (msh) and Epidermal growth factor receptor (Egfr). To investigate the evolution of neural pattern formation we identified and determined the expression patterns of Tribolium vnd, ind and msh, and find they are expressed in the medial, intermediate and lateral columns of the developing CNS, respectively, in patterns similar, but not identical, to their Drosophila orthologs. The pattern of Egfr activity suggests that the genetic regulatory mechanisms that initiate Tc-vnd expression are similar in Drosophila and Tribolium, whereas those that initiate Tc-ind have diverged. RNAi analyses of gene function show that Tc-vnd and Tc-ind promote the formation of medial and intermediate column neural precursors and that vnd-mediated repression of ind establishes the boundary between the medial and intermediate columns. These data suggest that columnar gene expression and function underlie neural pattern formation in Drosophila, Tribolium, and potentially all insects, but that subtle spatiotemporal differences in expression of these genes may produce species-specific morphological differences. Developmental Biology 279: 491-500 , Mar-2005
Abstracts for Journals and Proceedings	Wheeler, SR and Skeath JB "achaete-scute Gene Expression During Neurogenesis in the Crustacean Triops longicaudatus" The achaete-scute (ac/sc) genes are a paradigm for understanding the genesis of neural pattern. Prior to this work ac/sc gene expression had been investigated in insects, chelicerates, and myriapods. This work identified two ac/sc genes from the branchiopod crustacean Triops longicaudatus. Triops longicaudatus achaete-scute homologs1 and 2 (Tl-ASH1 and Tl-ASH2) exhibit dynamic and distinct expression profiles during Triops neurogenesis. Tl-ASH1 expression initiates in nearly all cells of the neurogenic region and subsequently in clusters of cells evenly spaced along the length of the developing limbs. In contrast, Tl-ASH2 initiates expression after Tl-ASH1. In the CNS, only a subset of Tl-ASH1 cells appears to express Tl-ASH2. Similarly, in the PNS individual Tl-ASH2 positive cells appear to arise from the clusters of Tl-ASH1 expressing cells. Shortly after activating Tl-ASH2 expression, these cells enlarge and divide. The expression dynamics of ac/sc genes in Triops parallel those observed in insects and contrasts with those found in chelicerates and myriapods. Gene Expr Patterns. 5 :695-700 , Oct-2005